

B5 General Relativity & Cosmology — Model Solutions, 2019

Question 1 — Equivalence principle and geodesic equation

(a) Equivalence principle and proper time

Problem. State the equivalence principle. In a free-falling lab with coordinates ξ^α , a particle obeys $d^2\xi^\alpha/d\tau^2 = 0$. Define the proper time τ .

Equivalence principle (strong form): at every spacetime event one can choose a local inertial (free-falling) frame in which the laws of physics take their special-relativistic form, and gravitational effects are locally indistinguishable from acceleration.

Free-falling frame: $\eta_{\alpha\beta} = \text{diag}(-1, +1, +1, +1)$. Invariant line element:

$$ds^2 = \eta_{\alpha\beta} d\xi^\alpha d\xi^\beta = -(d\xi^0)^2 + (d\boldsymbol{\xi})^2.$$

For a timelike worldline, $ds^2 < 0$. Define proper time as the interval read by a clock comoving with the particle:

$$c^2 d\tau^2 \equiv -ds^2 = -\eta_{\alpha\beta} d\xi^\alpha d\xi^\beta.$$

Divide by c^2 :

$$\boxed{d\tau^2 = -\frac{1}{c^2} \eta_{\alpha\beta} d\xi^\alpha d\xi^\beta.}$$

(b) Transformation to general coordinates

Problem. Starting from $d^2\xi^\alpha/d\tau^2 = 0$, derive the geodesic equation in coordinates x^α and identify the symmetry of Γ .

Chain rule on $\xi^\alpha = \xi^\alpha(x^\mu)$:

$$\frac{d\xi^\alpha}{d\tau} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{dx^\mu}{d\tau}.$$

Differentiate w.r.t. τ , applying product rule:

$$\frac{d^2\xi^\alpha}{d\tau^2} = \frac{d}{d\tau} \left(\frac{\partial \xi^\alpha}{\partial x^\mu} \right) \frac{dx^\mu}{d\tau} + \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{d^2x^\mu}{d\tau^2}.$$

Chain rule on $\partial \xi^\alpha / \partial x^\mu$ along the curve:

$$\frac{d}{d\tau} \left(\frac{\partial \xi^\alpha}{\partial x^\mu} \right) = \frac{\partial^2 \xi^\alpha}{\partial x^\nu \partial x^\mu} \frac{dx^\nu}{d\tau}.$$

Combine and set equal to zero:

$$\frac{\partial \xi^\alpha}{\partial x^\mu} \frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Multiply by $\partial x^\lambda / \partial \xi^\alpha$ (sum on α):

$$\frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{d^2 x^\mu}{d\tau^2} + \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Apply $(\partial x^\lambda / \partial \xi^\alpha)(\partial \xi^\alpha / \partial x^\mu) = \delta_\mu^\lambda$:

$$\delta_\mu^\lambda \frac{d^2 x^\mu}{d\tau^2} + \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Contract the Kronecker delta:

$$\frac{d^2 x^\lambda}{d\tau^2} + \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Identify the connection:

$$\Gamma_{\mu\nu}^\lambda \equiv \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}.$$

Geodesic equation:

$$\boxed{\frac{d^2 x^\lambda}{d\tau^2} + \Gamma_{\mu\nu}^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.}$$

Partial derivatives commute, hence $\Gamma_{\mu\nu}^\lambda = \Gamma_{\nu\mu}^\lambda$ (symmetric in lower indices).

(c) Metric compatibility relation

Problem. With $g_{\mu\nu} = \eta_{\alpha\beta}(\partial \xi^\alpha / \partial x^\mu)(\partial \xi^\beta / \partial x^\nu)$, show $\partial_\lambda g_{\mu\nu} = g_{\rho\nu} \Gamma_{\lambda\mu}^\rho + g_{\mu\rho} \Gamma_{\lambda\nu}^\rho$.

Invariant line element:

$$ds^2 = \eta_{\alpha\beta} d\xi^\alpha d\xi^\beta = g_{\mu\nu} dx^\mu dx^\nu, \quad g_{\mu\nu} = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu}.$$

Differentiate $g_{\mu\nu}$ w.r.t. x^λ , using product rule:

$$\frac{\partial g_{\mu\nu}}{\partial x^\lambda} = \eta_{\alpha\beta} \frac{\partial^2 \xi^\alpha}{\partial x^\lambda \partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} + \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial^2 \xi^\beta}{\partial x^\lambda \partial x^\nu}.$$

Invert the connection definition: multiply $\Gamma_{\mu\nu}^\lambda = (\partial x^\lambda / \partial \xi^\alpha)(\partial^2 \xi^\alpha / \partial x^\mu \partial x^\nu)$ by $\partial \xi^\beta / \partial x^\lambda$ and apply $(\partial \xi^\beta / \partial x^\lambda)(\partial x^\lambda / \partial \xi^\alpha) = \delta_\alpha^\beta$:

$$\frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} = \Gamma_{\mu\nu}^\rho \frac{\partial \xi^\alpha}{\partial x^\rho}.$$

Substitute this identity (with index labels (λ, μ) and (λ, ν)):

$$\frac{\partial g_{\mu\nu}}{\partial x^\lambda} = \eta_{\alpha\beta} \Gamma_{\lambda\mu}^\rho \frac{\partial \xi^\alpha}{\partial x^\rho} \frac{\partial \xi^\beta}{\partial x^\nu} + \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} \Gamma_{\lambda\nu}^\rho \frac{\partial \xi^\beta}{\partial x^\rho}.$$

Recognise $g_{\rho\nu}$ and $g_{\mu\rho}$ from the metric definition:

$$\boxed{\frac{\partial g_{\mu\nu}}{\partial x^\lambda} = g_{\rho\nu} \Gamma_{\lambda\mu}^\rho + g_{\mu\rho} \Gamma_{\lambda\nu}^\rho.}$$

(d) Christoffel formula

Problem. Use the symmetry of Γ to express $\Gamma_{\mu\lambda}^\sigma$ in terms of $g_{\mu\nu}$.

Restate ():

$$\partial_\lambda g_{\mu\nu} = g_{\rho\nu} \Gamma_{\lambda\mu}^\rho + g_{\mu\rho} \Gamma_{\lambda\nu}^\rho.$$

Cyclic permutation $\lambda \rightarrow \mu \rightarrow \nu \rightarrow \lambda$:

$$\partial_\mu g_{\nu\lambda} = g_{\rho\lambda} \Gamma_{\mu\nu}^\rho + g_{\nu\rho} \Gamma_{\mu\lambda}^\rho.$$

Cyclic permutation once more:

$$\partial_\nu g_{\lambda\mu} = g_{\rho\mu} \Gamma_{\nu\lambda}^\rho + g_{\lambda\rho} \Gamma_{\nu\mu}^\rho.$$

Form $\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}$:

$$\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu} = g_{\rho\lambda} \Gamma_{\mu\nu}^\rho + g_{\nu\rho} \Gamma_{\mu\lambda}^\rho + g_{\rho\mu} \Gamma_{\nu\lambda}^\rho + g_{\lambda\rho} \Gamma_{\nu\mu}^\rho - g_{\rho\nu} \Gamma_{\lambda\mu}^\rho - g_{\mu\rho} \Gamma_{\lambda\nu}^\rho.$$

Apply $\Gamma_{\mu\nu}^\rho = \Gamma_{\nu\mu}^\rho$ to cancel the mixed pairs:

$$g_{\nu\rho} \Gamma_{\mu\lambda}^\rho - g_{\mu\rho} \Gamma_{\lambda\nu}^\rho = 0, \quad g_{\rho\mu} \Gamma_{\nu\lambda}^\rho - g_{\rho\nu} \Gamma_{\lambda\mu}^\rho = 0.$$

Surviving terms (both equal $g_{\rho\lambda} \Gamma_{\mu\nu}^\rho$ by symmetry):

$$\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu} = 2 g_{\rho\lambda} \Gamma_{\mu\nu}^\rho.$$

Multiply by $\frac{1}{2} g^{\sigma\lambda}$ and use $g^{\sigma\lambda} g_{\rho\lambda} = \delta_\rho^\sigma$:

$$\frac{1}{2} g^{\sigma\lambda} (\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}) = \Gamma_{\mu\nu}^\sigma.$$

Relabel $\nu \rightarrow \lambda, \lambda \rightarrow \nu$ to match the requested form:

$$\Gamma_{\mu\lambda}^\sigma = \frac{1}{2} g^{\nu\sigma} \left(\frac{\partial g_{\mu\nu}}{\partial x^\lambda} + \frac{\partial g_{\lambda\nu}}{\partial x^\mu} - \frac{\partial g_{\lambda\mu}}{\partial x^\nu} \right).$$

(e) Covariant derivative

Problem. Define $U^\lambda = dx^\lambda/d\tau$. Use the geodesic equation to define the covariant derivative; comment on its rank and significance.

Geodesic equation rewritten using U^λ :

$$\frac{dU^\lambda}{d\tau} + \Gamma_{\mu\nu}^\lambda U^\mu U^\nu = 0.$$

Apply chain rule along the curve: $dU^\lambda/d\tau = (\partial_\nu U^\lambda)(dx^\nu/d\tau) = U^\nu \partial_\nu U^\lambda$:

$$U^\nu \partial_\nu U^\lambda + \Gamma_{\mu\nu}^\lambda U^\mu U^\nu = 0.$$

Factor U^ν :

$$U^\nu \left(\partial_\nu U^\lambda + \Gamma_{\mu\nu}^\lambda U^\mu \right) = 0.$$

Identify the bracket as the covariant derivative:

$$\nabla_\nu U^\lambda \equiv \partial_\nu U^\lambda + \Gamma_{\mu\nu}^\lambda U^\mu.$$

Significance. $\nabla_\nu U^\lambda$ corrects the partial derivative for the rotation/stretching of the coordinate basis induced by curvature, so that geodesic motion ($\nabla_U U = 0$) is the curved-space analogue of Newton's first law: a free particle parallel-transport its own four-velocity.

Tensor character. Under $x^\mu \rightarrow x'^\mu$, transform $\partial_\nu U^\lambda$ using product rule on the Jacobian:

$$\partial_\nu U^\lambda \rightarrow \frac{\partial x^\beta}{\partial x'^\nu} \left(\frac{\partial x'^\lambda}{\partial x^\alpha} \partial_\beta U^\alpha + U^\alpha \frac{\partial^2 x'^\lambda}{\partial x^\alpha \partial x^\beta} \right).$$

Transform $\Gamma_{\mu\nu}^\lambda U^\mu$ (the inhomogeneous transformation law of Γ contributes a second-derivative piece with opposite sign):

$$\Gamma_{\mu\nu}^\lambda U^\mu \rightarrow \frac{\partial x'^\lambda}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x'^\nu} \Gamma_{\gamma\beta}^\alpha U^\gamma - \frac{\partial x^\beta}{\partial x'^\nu} U^\alpha \frac{\partial^2 x'^\lambda}{\partial x^\alpha \partial x^\beta}.$$

Add: the non-tensorial second-derivative pieces cancel:

$$\nabla_\nu U^\lambda \rightarrow \frac{\partial x'^\lambda}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x'^\nu} \nabla_\beta U^\alpha.$$

Hence $\nabla_\nu U^\lambda$ transforms as a $(1, 1)$ mixed rank-2 tensor (one upper Jacobian, one lower inverse Jacobian) — required so that $\nabla_U U = 0$ has covariant meaning.

Question 2 — Schwarzschild geodesics and orbits

(a) Geodesic equations from the Lagrangian

Problem. Write the geodesic equations for the Schwarzschild line element using the Lagrange procedure; explain the θ equation; relate p^α to the geodesics.

Effective Lagrangian (one-half line element):

$$\mathcal{L} = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = \frac{1}{2} \left[- \left(1 - \frac{2GM}{c^2 r} \right) c^2 \dot{t}^2 + \left(1 - \frac{2GM}{c^2 r} \right)^{-1} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2 \right].$$

Euler–Lagrange equations $\frac{d}{d\tau} \partial_{\dot{x}^\mu} \mathcal{L} - \partial_{x^\mu} \mathcal{L} = 0$:

$$\frac{d}{d\tau} \left[\left(1 - \frac{2GM}{c^2 r} \right) c^2 \dot{t} \right] = 0,$$

$$\frac{d}{d\tau} \left[\left(1 - \frac{2GM}{c^2 r} \right)^{-1} \dot{r} \right] - \frac{1}{2} \left[- \frac{2GM}{c^2 r^2} c^2 \dot{t}^2 - \frac{2GM}{c^2 r^2} \left(1 - \frac{2GM}{c^2 r} \right)^{-2} \dot{r}^2 + 2r \dot{\theta}^2 + 2r \sin^2 \theta \dot{\phi}^2 \right] = 0,$$

$$\frac{d}{d\tau} \left[r^2 \dot{\theta} \right] - r^2 \sin \theta \cos \theta \dot{\phi}^2 = 0,$$

$$\frac{d}{d\tau} \left[r^2 \sin^2 \theta \dot{\phi} \right] = 0.$$

Spherical symmetry: orient axes so initial conditions are $\theta = \pi/2$, $\dot{\theta} = 0$. Insert into the θ -equation:

$$\frac{d}{d\tau} (r^2 \cdot 0) - r^2 \sin(\pi/2) \cos(\pi/2) \dot{\phi}^2 = 0.$$

Use $\cos(\pi/2) = 0$:

$$\ddot{\theta} = 0.$$

Hence $\theta(\tau) = \pi/2$ is preserved identically and the orbit lies in the equatorial plane.

Four-momentum. For a massive particle $p^\alpha = m dx^\alpha/d\tau$; multiplying the geodesic equation by m gives the parallel-transport law $p^\nu \nabla_\nu p^\alpha = 0$, i.e. free particles follow geodesic curves with their four-momentum tangent to the geodesic.

(b) Affine parameter and constants of motion

Problem. When is τ an appropriate affine parameter? Identify the conserved quantities k and h from the t and ϕ equations.

Proper time τ is affine only for *timelike* (massive) geodesics, where $d\tau^2 = -ds^2/c^2 > 0$. For *null* (massless) geodesics $d\tau = 0$, so pick any affine parameter λ along which $p^\alpha = dx^\alpha/d\lambda$ and the geodesic equation $dp^\alpha/d\lambda + \Gamma_{\mu\nu}^\alpha p^\mu (dx^\nu/d\lambda) = 0$ holds.

Integrate the t -equation (Lagrangian independent of t , so $\partial_t \mathcal{L}$ conserved):

$$\left(1 - \frac{2GM}{c^2 r}\right) c^2 \dot{t} = \text{const} \equiv c^2 k.$$

Integrate the ϕ -equation in the equatorial plane ($\sin^2 \theta = 1$):

$$r^2 \dot{\phi} = \text{const} \equiv h.$$

Lower the index using $p_\alpha = g_{\alpha\beta} p^\beta = m g_{\alpha\beta} \dot{x}^\beta$. For $\alpha = 0$ use $g_{tt} = -(1 - 2GM/c^2 r)c^2$:

$$p_0 = -m \left(1 - \frac{2GM}{c^2 r}\right) c^2 \dot{t} = -mc^2 k.$$

For $\alpha = 3$ use $g_{\phi\phi} = r^2 \sin^2 \theta$:

$$p_3 = m r^2 \sin^2 \theta \dot{\phi} = m h.$$

Physical meaning. $k = -p_0/(mc^2)$ is the conserved *energy per unit rest mass* measured at infinity (associated with Killing vector ∂_t via time-translation symmetry); $h = p_3/m$ is the conserved *specific angular momentum* (Killing vector ∂_ϕ via rotational symmetry).

(c) Energy equation

Problem. From $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = c^2$ derive the radial energy equation.

Divide $-c^2 d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu$ by $d\tau^2$, set $\dot{\theta} = 0$, $\theta = \pi/2$:

$$-\left(1 - \frac{2GM}{c^2 r}\right) c^2 \dot{t}^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2 = -c^2.$$

Solve for $\dot{t}$ from $k = (1 - 2GM/c^2 r)\dot{t}$:

$$\dot{t} = \frac{k}{1 - 2GM/(c^2 r)}.$$

Substitute $\dot{t}$ and $\dot{\phi} = h/r^2$:

$$-\left(1 - \frac{2GM}{c^2 r}\right) \frac{c^2 k^2}{(1 - 2GM/c^2 r)^2} + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} \dot{r}^2 + \frac{h^2}{r^2} = -c^2.$$

Simplify the first term:

$$-\frac{c^2 k^2}{1 - 2GM/(c^2 r)} + \frac{\dot{r}^2}{1 - 2GM/(c^2 r)} + \frac{h^2}{r^2} = -c^2.$$

Multiply through by $(1 - 2GM/(c^2 r))$:

$$-c^2 k^2 + \dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2GM}{c^2 r}\right) = -c^2 \left(1 - \frac{2GM}{c^2 r}\right).$$

Expand the right-hand side:

$$-c^2 k^2 + \dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2GM}{c^2 r}\right) = -c^2 + \frac{2GM}{r}.$$

Rearrange (move $-c^2 k^2$ and $2GM/r$ across):

$$\boxed{\dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2GM}{c^2 r}\right) - \frac{2GM}{r} = c^2 (k^2 - 1).}$$

(d) Orbit equation and relativistic correction

Problem. Using () and $r^2\dot{\phi} = h$ derive $d^2u/d\phi^2 + u = GM/h^2 + 3GMu^2/c^2$ with $u = 1/r$.

Change variable $u = 1/r$, so $r = 1/u$ and $dr/d\phi = -u^{-2}du/d\phi$. Use chain rule with $\dot{\phi} = h/r^2 = hu^2$:

$$\dot{r} = \frac{dr}{d\phi} \dot{\phi} = -\frac{1}{u^2} \frac{du}{d\phi} \cdot hu^2.$$

Cancel u^2 :

$$\dot{r} = -h \frac{du}{d\phi}.$$

Square:

$$\dot{r}^2 = h^2 \left(\frac{du}{d\phi} \right)^2.$$

Substitute $\dot{r}^2$ and $1/r = u$ into ():

$$h^2 \left(\frac{du}{d\phi} \right)^2 + h^2 u^2 \left(1 - \frac{2GMu}{c^2} \right) - 2GMu = c^2(k^2 - 1).$$

Expand the bracket on the second term:

$$h^2 \left(\frac{du}{d\phi} \right)^2 + h^2 u^2 - \frac{2GMh^2}{c^2} u^3 - 2GMu = c^2(k^2 - 1).$$

Differentiate w.r.t. ϕ (RHS is constant):

$$2h^2 \frac{du}{d\phi} \frac{d^2u}{d\phi^2} + 2h^2 u \frac{du}{d\phi} - \frac{6GMh^2}{c^2} u^2 \frac{du}{d\phi} - 2GM \frac{du}{d\phi} = 0.$$

Factor out $2du/d\phi$:

$$2 \frac{du}{d\phi} \left[h^2 \frac{d^2u}{d\phi^2} + h^2 u - \frac{3GMh^2}{c^2} u^2 - GM \right] = 0.$$

Non-circular orbit: $du/d\phi \neq 0$ so the bracket vanishes:

$$h^2 \frac{d^2u}{d\phi^2} + h^2 u - \frac{3GMh^2}{c^2} u^2 - GM = 0.$$

Divide by h^2 :

$$\frac{d^2u}{d\phi^2} + u - \frac{3GM}{c^2} u^2 - \frac{GM}{h^2} = 0.$$

Rearrange to the requested form:

$$\boxed{\frac{d^2u}{d\phi^2} + u = \frac{GM}{h^2} + \frac{3GM}{c^2} u^2.}$$

The Newtonian equation has only the constant source GM/h^2 . The relativistic correction is the term $3GMu^2/c^2$. Its ratio to the Newtonian term is $3h^2u^2/c^2 = 3v_\perp^2/c^2$ where $v_\perp = hu$, so it is significant when $v \sim c$, i.e. for tight orbits at $r \sim r_s = 2GM/c^2$ (e.g. Mercury's perihelion advance, S2 around Sgr A*).

(e) Effective potential in GR

Problem. Cast the energy equation in Newtonian form $\frac{1}{2}(\dot{r})^2 + V_{\text{eff}}(r) = E$.

In the orbiting observer's frame the affine parameter is τ , so dr/dt is replaced by $dr/d\tau = \dot{r}$. Multiply () by $\frac{1}{2}$:

$$\frac{1}{2}\dot{r}^2 + \frac{1}{2}\left[\frac{h^2}{r^2}\left(1 - \frac{2GM}{c^2 r}\right) - \frac{2GM}{r}\right] = \frac{1}{2}c^2(k^2 - 1).$$

Expand the inner bracket:

$$\frac{1}{2}\dot{r}^2 + \frac{h^2}{2r^2} - \frac{GMh^2}{c^2 r^3} - \frac{GM}{r} = \frac{1}{2}c^2(k^2 - 1).$$

Match to the form $\frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) = E$:

$$V_{\text{eff}}(r) = -\frac{GM}{r} + \frac{h^2}{2r^2} - \frac{GMh^2}{c^2 r^3}, \quad E = \frac{1}{2}c^2(k^2 - 1).$$

The first two terms reproduce Newtonian gravity + centrifugal barrier; the $1/r^3$ term is purely relativistic.

(f) Circular orbits

Problem. Show circular-orbit extrema occur at $r = (h/2GM)\left(h \pm \sqrt{h^2 - 12G^2M^2/c^2}\right)$.

Circular orbit condition: $dV_{\text{eff}}/dr = 0$. Differentiate () term by term:

$$\frac{dV_{\text{eff}}}{dr} = \frac{GM}{r^2} - \frac{h^2}{r^3} + \frac{3GMh^2}{c^2 r^4}.$$

Set to zero:

$$\frac{GM}{r^2} - \frac{h^2}{r^3} + \frac{3GMh^2}{c^2 r^4} = 0.$$

Multiply by r^4 :

$$GM r^2 - h^2 r + \frac{3GMh^2}{c^2} = 0.$$

Solve the quadratic in r via the standard formula:

$$r = \frac{h^2 \pm \sqrt{h^4 - 4 \cdot GM \cdot 3GMh^2/c^2}}{2GM}.$$

Simplify under the surd:

$$r = \frac{h^2 \pm \sqrt{h^4 - 12G^2M^2h^2/c^2}}{2GM}.$$

Factor h^2 inside the surd:

$$r = \frac{h^2 \pm h\sqrt{h^2 - 12G^2M^2/c^2}}{2GM}.$$

Pull out the common factor h in the numerator:

$$r = \frac{h}{2GM} \left(h \pm \sqrt{h^2 - \frac{12G^2M^2}{c^2}} \right).$$

The $+$ root is the stable minimum of V_{eff} (regular orbit), the $-$ root is the unstable maximum (light-ring / unstable circular orbit). Real solutions require $h^2 \geq 12G^2M^2/c^2$; equality gives the innermost stable circular orbit (ISCO) at $r = 6GM/c^2$.

Question 3 — Binary pulsar and gravitational radiation

(a) Why a GR expert should notice

Problem. *Pulsar in a binary, $T = 0.102\,251\,562\,48$ day. Why does this matter for GR?*

A binary system containing a pulsar provides a clean clock embedded in a strong-field, dynamical spacetime: the orbit decays via emission of gravitational waves, and the highly accurate pulse timing lets one measure $\dot{T}$ at the level needed to test the quadrupole formula of GR (Hulse–Taylor / double pulsar). It is one of very few systems testing the radiative sector of Einstein’s theory.

(b) Orbital radius

Problem. $T_0 = 0.102\,251\,562\,48$ day, $\dot{T} \approx -10^{-9}$ day/1000 day, $m_1 = m_2 = 1.4M_\odot$, *circular orbit. Find a .*

Convert period to SI:

$$T = 0.10225156248 \times 86400 \text{ s} = 8834.535 \text{ s}.$$

Apply Kepler’s third law (total mass $M_{\text{tot}} = 2m_p = 2.8 M_\odot$, $GM_\odot = 1.327 \times 10^{20} \text{ m}^3 \text{ s}^{-2}$):

$$a^3 = \frac{GM_{\text{tot}}T^2}{4\pi^2}.$$

Evaluate GM_{tot} :

$$GM_{\text{tot}} = 2.8 \times 1.327 \times 10^{20} \text{ m}^3 \text{ s}^{-2} = 3.716 \times 10^{20} \text{ m}^3 \text{ s}^{-2}.$$

Evaluate T^2 :

$$T^2 = (8834.535 \text{ s})^2 = 7.805 \times 10^7 \text{ s}^2.$$

Multiply:

$$GM_{\text{tot}}T^2 = 3.716 \times 10^{20} \text{ m}^3 \text{ s}^{-2} \times 7.805 \times 10^7 \text{ s}^2 = 2.901 \times 10^{28} \text{ m}^3.$$

Divide by $4\pi^2$:

$$a^3 = \frac{2.901 \times 10^{28} \text{ m}^3}{4\pi^2} = 7.346 \times 10^{26} \text{ m}^3.$$

Take the cube root:

$$a = (7.346 \times 10^{26} \text{ m}^3)^{1/3} \approx 9.02 \times 10^8 \text{ m}.$$

Compare to $R_\odot = 6.96 \times 10^8 \text{ m}$:

$$a \approx 9.02 \times 10^8 \text{ m} \approx 1.3 R_\odot.$$

(c) Definitions in the quadrupole formula

Problem. *Define $\bar{h}^{ij}$ and t' in $\bar{h}^{ij} = (2G/c^6 r) d^2 I^{ij} / dt'^2$.*

$\bar{h}^{\mu\nu}$ is the *trace-reversed* metric perturbation in the harmonic gauge:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad \bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h, \quad h \equiv \eta^{\mu\nu} h_{\mu\nu}.$$

The harmonic-gauge condition $\partial_\mu \bar{h}^{\mu\nu} = 0$ reduces the linearised Einstein equation to $\square \bar{h}^{\mu\nu} = (16\pi G/c^4) T^{\mu\nu}$ (sign consistent with the paper’s $-8\pi G/c^4$ convention for the Einstein equation), whose retarded solution gives the quadrupole formula.

t' is the *retarded time* at the source:

$$t' = t - r/c,$$

the time of emission for a wave received at the field point $(t, \mathbf{r})$.

(d) Meaning of x^i, x^j

Problem. What do x^i, x^j in $I^{ij} = Mc^2 x^i x^j$ represent, given that pulsars are point-like?

Pulsars are ~ 10 km across against orbital separations of $\sim 10^9$ m, so each behaves as a point mass and the internal structure does not contribute to I^{ij} . For two equal masses $M_p = M/2$ on a circular orbit, each pulsar sits at $\pm \mathbf{x}(t')$ with $|\mathbf{x}| = a/2$ on opposite sides of the centre of mass. The standard quadrupole sum is

$$I^{ij} = \sum_{A=1,2} M_p c^2 x_A^i x_A^j = 2M_p c^2 x^i x^j = M c^2 x^i x^j.$$

Hence x^i are the *Cartesian components of one pulsar's orbital position vector about the centre of mass*, of magnitude $a/2$. They label the macroscopic orbital trajectory, not internal pulsar coordinates.

(e) GW frequency and waveform

Problem. Find f_{GW} to 2 s.f.; write $\bar{h}^{ij}(t')$.

Orbital angular frequency:

$$\omega_{\text{orb}} = \frac{2\pi}{T} = \frac{2\pi}{8834.535 \text{ s}} = 7.111 \times 10^{-4} \text{ rad/s}.$$

Orbital frequency:

$$f_{\text{orb}} = \frac{1}{T} = \frac{1}{8834.535 \text{ s}} = 1.132 \times 10^{-4} \text{ Hz}.$$

Quadrupole radiation: $f_{\text{GW}} = 2f_{\text{orb}}$:

$$f_{\text{GW}} = 2 \times 1.132 \times 10^{-4} \text{ Hz} = 2.26 \times 10^{-4} \text{ Hz}.$$

Round to 2 s.f.:

$f_{\text{GW}} \approx 2.3 \times 10^{-4} \text{ Hz}.$

Waveform. Place orbit in xy -plane. Pulsar 1 sits at $\mathbf{x} = (a/2)(\cos \omega t', \sin \omega t', 0)$; pulsar 2 at $-\mathbf{x}$. Sum quadrupole contributions:

$$I^{ij} = M_p c^2 x^i x^j + M_p c^2 (-x)^i (-x)^j = 2M_p c^2 x^i x^j = M c^2 x^i x^j,$$

matching the formula in the question (with $M = 2M_p$). Take $\omega \equiv \omega_{\text{orb}}$ and compute components:

$$x^1(t') = (a/2) \cos(\omega t'), \quad x^2(t') = (a/2) \sin(\omega t'), \quad x^3 = 0.$$

Hence:

$$I^{11} = M c^2 (a/2)^2 \cos^2(\omega t'),$$

$$I^{22} = M c^2 (a/2)^2 \sin^2(\omega t'),$$

$$I^{12} = M c^2 (a/2)^2 \sin(\omega t') \cos(\omega t').$$

Apply $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$:

$$I^{11} = \frac{1}{2} M c^2 (a/2)^2 [1 + \cos(2\omega t')].$$

Apply $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$:

$$I^{22} = \frac{1}{2} M c^2 (a/2)^2 [1 - \cos(2\omega t')].$$

Apply $\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$:

$$I^{12} = \frac{1}{2} M c^2 (a/2)^2 \sin(2\omega t').$$

Differentiate twice w.r.t. t' (constants drop; $d^2/dt'^2[\cos 2\omega t'] = -4\omega^2 \cos 2\omega t'$):

$$\ddot{I}^{11} = -\frac{1}{2}Mc^2(a/2)^2 \cdot 4\omega^2 \cos(2\omega t').$$

Collect $(a/2)^2 \cdot 4 = a^2$:

$$\ddot{I}^{11} = -\frac{1}{2}Mc^2a^2\omega^2 \cos(2\omega t').$$

Similarly $\ddot{I}^{22}$ (sign flips because $1 - \cos 2\omega t'$):

$$\ddot{I}^{22} = +\frac{1}{2}Mc^2a^2\omega^2 \cos(2\omega t').$$

And $\ddot{I}^{12}$ (second derivative of $\sin 2\omega t'$ is $-4\omega^2 \sin 2\omega t'$):

$$\ddot{I}^{12} = -\frac{1}{2}Mc^2a^2\omega^2 \sin(2\omega t').$$

Insert into $\bar{h}^{ij} = (2G/c^6r)\ddot{I}^{ij}$:

$$\bar{h}^{11} = \frac{2G}{c^6r} \cdot \left(-\frac{1}{2}Mc^2a^2\omega^2 \cos 2\omega t'\right).$$

Cancel the factor of 2 and reduce $c^2/c^6 = 1/c^4$:

$$\bar{h}^{11} = -\frac{GMa^2\omega^2}{c^4r} \cos(2\omega t').$$

Similarly for the remaining components:

$$\bar{h}^{22} = +\frac{GMa^2\omega^2}{c^4r} \cos(2\omega t'), \quad \bar{h}^{12} = \bar{h}^{21} = -\frac{GMa^2\omega^2}{c^4r} \sin(2\omega t').$$

Collect into a matrix:

$$\boxed{\bar{h}^{ij}(t') = \frac{GMa^2\omega^2}{c^4r} \begin{pmatrix} -\cos 2\omega t' & -\sin 2\omega t' & 0 \\ -\sin 2\omega t' & +\cos 2\omega t' & 0 \\ 0 & 0 & 0 \end{pmatrix},}$$

oscillating at angular frequency $2\omega_{\text{orb}}$, hence frequency 2.3×10^{-4} Hz.

(f) Why $\dot{T}$ tests GR

Problem. *Why does measuring $\dot{T}$ test general relativity?*

Energy radiated as gravitational waves, summed over the two polarisations and integrated over the sphere, is the Einstein quadrupole luminosity formula (with reduced quadrupole $Q^{ij} = Mx^i x^j - \frac{1}{3}\delta^{ij}\delta_{kl}Mx^k x^l$, i.e. I^{ij}/c^2 minus trace):

$$P_{\text{GW}} = \frac{G}{5c^5} \langle \ddot{Q}_{ij} \ddot{Q}^{ij} \rangle.$$

For a circular binary of masses m_1, m_2 and separation a this evaluates to

$$P_{\text{GW}} = \frac{32}{5} \frac{G^4}{c^5} \frac{(m_1 m_2)^2 (m_1 + m_2)}{a^5}.$$

This luminosity drains the orbital binding energy $E_{\text{orb}} = -Gm_1 m_2/(2a)$. Energy balance $\dot{E}_{\text{orb}} = -P_{\text{GW}}$ shrinks a :

$$\dot{a} = -\frac{64}{5} \frac{G^3}{c^5} \frac{m_1 m_2 (m_1 + m_2)}{a^3}.$$

Kepler's third law $T^2 \propto a^3/(m_1 + m_2)$ gives $\dot{T}/T = \frac{3}{2}\dot{a}/a$, so

$$\dot{T} = -\frac{192\pi}{5c^5} \left(\frac{2\pi G}{T} \right)^{5/3} \frac{m_1 m_2}{(m_1 + m_2)^{1/3}}.$$

$\dot{T}$ is therefore predicted with *no free parameters* once m_1, m_2 are known. The slight eccentricity permits independent measurement of post-Keplerian parameters (periastron advance, Shapiro delay, gravitational redshift) which over-determine m_1, m_2 . Comparing predicted $\dot{T}$ to the measured one is thus a direct, parameter-free test of GR's radiative quadrupole sector — famously passed by Hulse–Taylor and by the double pulsar to better than 0.1%.

Question 4 — Cosmological constant and the FRW universe

(a) Bianchi identities

Problem. *State the contracted Bianchi identities and verify that the modified Einstein equation respects them. Why is this physically significant?*

Contracted Bianchi identity (Einstein tensor is divergence-free):

$$\nabla^\mu (R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R) = 0.$$

Take the covariant divergence ∇^μ of equation (2):

$$\nabla^\mu (R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R) + \Lambda \nabla^\mu g_{\mu\nu} = -\frac{8\pi G}{c^4} \nabla^\mu T_{\mu\nu}.$$

Apply Bianchi (first term vanishes) and metric compatibility $\nabla^\mu g_{\mu\nu} = 0$ (second term vanishes):

$$0 + 0 = -\frac{8\pi G}{c^4} \nabla^\mu T_{\mu\nu}.$$

Conclude:

$$\nabla^\mu T_{\mu\nu} = 0.$$

Significance. The modified equation enforces local conservation of energy–momentum automatically. Adding $\Lambda g_{\mu\nu}$ does not spoil this because $g_{\mu\nu}$ is covariantly constant; any other proposed addition would have to share that property.

(b) Four-velocity of the cosmic fluid

Problem. *Define U^μ in $T_{\mu\nu} = P g_{\mu\nu} + (\rho + P/c^2) U_\mu U_\nu$. Give U^λ, U_λ at $\lambda = t, r$ in the FRW comoving frame.*

$U^\mu = dx^\mu/d\tau$ is the four-velocity of a fluid element measured in its own proper time, normalised $g_{\mu\nu} U^\mu U^\nu = -c^2$.

Comoving frame: fluid at rest in spatial coordinates, $dr = d\theta = d\phi = 0$. Reduce line element:

$$-c^2 d\tau^2 = -c^2 dt^2.$$

Solve for $dt/d\tau$ (take positive root):

$$\frac{dt}{d\tau} = 1.$$

Identify the four-velocity components:

$$U^t = \frac{dt}{d\tau} = 1, \quad U^r = \frac{dr}{d\tau} = 0.$$

Read off the diagonal metric components: $g_{tt} = -c^2$, $g_{rr} = R^2(t)/(1 - r^2/a^2)$:

$$U_t = g_{tt}U^t = (-c^2)(1) = -c^2.$$

$$U_r = g_{rr}U^r = \frac{R^2}{1 - r^2/a^2} \cdot 0 = 0.$$

Collect:

$$\boxed{U^t = 1, \quad U^r = 0; \quad U_t = -c^2, \quad U_r = 0.}$$

(c) Effective fluid with Λ

Problem. Show that with $\rho \rightarrow \rho'$, $P \rightarrow P'$ the modified equation reduces to the original; evaluate ρ_V and interpret.

Form the primed stress tensor with $\rho \rightarrow \rho'$, $P \rightarrow P'$:

$$T'_{\mu\nu} = P' g_{\mu\nu} + (\rho' + P'/c^2)U_\mu U_\nu.$$

Substitute $\rho' = \rho + \rho_V$ and $P' = P - \rho_V c^2$:

$$T'_{\mu\nu} = (P - \rho_V c^2) g_{\mu\nu} + [(\rho + \rho_V) + (P - \rho_V c^2)/c^2]U_\mu U_\nu.$$

Simplify the bracket: the $+\rho_V$ and $-\rho_V c^2/c^2$ cancel:

$$(\rho + \rho_V) + (P - \rho_V c^2)/c^2 = \rho + P/c^2.$$

Hence:

$$T'_{\mu\nu} = (P - \rho_V c^2) g_{\mu\nu} + (\rho + P/c^2)U_\mu U_\nu.$$

Split using $T_{\mu\nu} = P g_{\mu\nu} + (\rho + P/c^2)U_\mu U_\nu$:

$$T'_{\mu\nu} = T_{\mu\nu} - \rho_V c^2 g_{\mu\nu}.$$

Insert into equation (1) with $T_{\mu\nu} \rightarrow T'_{\mu\nu}$:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\frac{8\pi G}{c^4}T'_{\mu\nu}.$$

Expand the RHS:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\frac{8\pi G}{c^4}T_{\mu\nu} + \frac{8\pi G}{c^2}\rho_V g_{\mu\nu}.$$

Rearrange the modified equation (2) into the same form:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\frac{8\pi G}{c^4}T_{\mu\nu} - \Lambda g_{\mu\nu}.$$

Match the coefficient of $g_{\mu\nu}$:

$$\frac{8\pi G}{c^2}\rho_V = -\Lambda.$$

Solve for ρ_V :

$$\boxed{\rho_V = -\frac{\Lambda c^2}{8\pi G}.$$

Interpretation. ρ_V is the energy density of the *vacuum*: a Lorentz-invariant perfect fluid with equation of state $P_V = -\rho_V c^2$, constant in space and time, indistinguishable from the cosmological-constant term in the field equations.

(d) **FRW identity** $\ddot{R}/R = H^2 + \dot{H}$

Problem. Prove $\ddot{R}/R = H^2 + \dot{H}$.

Definition $H = \dot{R}/R$ implies $\dot{R} = HR$. Differentiate w.r.t. t :

$$\ddot{R} = \dot{H}R + H\dot{R}.$$

Substitute $\dot{R} = HR$:

$$\ddot{R} = \dot{H}R + H^2R.$$

Divide by R :

$$\boxed{\frac{\ddot{R}}{R} = H^2 + \dot{H}.}$$

(e) **Differential equation for H**

Problem. For a flat Λ -matter universe show $2\dot{H} + 3H^2 = \Lambda c^2$.

For non-relativistic matter, $p = 0$. The two Friedmann equations reduce to

$$\frac{\ddot{R}}{R} = -\frac{4\pi G}{3}\rho + \frac{1}{3}\Lambda c^2,$$

$$H^2 = \frac{8\pi G}{3}\rho + \frac{1}{3}\Lambda c^2 \quad (k = 0).$$

Eliminate ρ . Multiply the second by $\frac{1}{2}$:

$$\frac{1}{2}H^2 = \frac{4\pi G}{3}\rho + \frac{1}{6}\Lambda c^2.$$

Add to the first:

$$\frac{\ddot{R}}{R} + \frac{1}{2}H^2 = \frac{1}{2}\Lambda c^2.$$

Use $(\dot{})$:

$$\dot{H} + H^2 + \frac{1}{2}H^2 = \frac{1}{2}\Lambda c^2.$$

Multiply by 2:

$$\boxed{2\dot{H} + 3H^2 = \Lambda c^2.}$$

(f) **Solving for $H(t)$**

Problem. If $H \propto 1/t$ at early times, show $H = A \coth(3At/2)$ with $A^2 = \Lambda c^2/3$, and comment on late-time behaviour.

Early-time check (matter domination, Λ negligible): $(\dot{})$ becomes $2\dot{H} + 3H^2 = 0$. Trial $H = c_1/t$ gives $-2c_1/t^2 + 3c_1^2/t^2 = 0$, hence $c_1 = 2/3$:

$$H_{\text{early}} = \frac{2}{3t} \propto \frac{1}{t}.$$

This matches the hypothesis and fixes the boundary condition $H \rightarrow \infty$ as $t \rightarrow 0^+$.

Define $A^2 \equiv \Lambda c^2/3$ and rearrange ():

$$2 \frac{dH}{dt} = \Lambda c^2 - 3H^2.$$

Factor the RHS:

$$2 \frac{dH}{dt} = 3(A^2 - H^2).$$

Separate variables:

$$\frac{2 dH}{A^2 - H^2} = 3 dt.$$

Integrate using the hint $\int dx/(p^2 - x^2) = (1/p) \coth^{-1}(x/p)$ with $p = A$, $x = H$:

$$\frac{2}{A} \coth^{-1}(H/A) = 3t + \text{const.}$$

Apply $H \rightarrow \infty$ as $t \rightarrow 0^+$, so $\coth^{-1}(H/A) \rightarrow 0$ and the constant is 0:

$$\frac{2}{A} \coth^{-1}(H/A) = 3t.$$

Solve for $\coth^{-1}(H/A)$:

$$\coth^{-1}(H/A) = \frac{3At}{2}.$$

Invert:

$$\boxed{H(t) = A \coth(3At/2), \quad A = \sqrt{\Lambda c^2/3}.$$

Late time. $\coth(3At/2) \rightarrow 1$ as $t \rightarrow \infty$:

$$H \rightarrow A = \sqrt{\Lambda c^2/3}.$$

The Hubble parameter approaches a constant, the universe enters de Sitter (exponential) expansion $R \propto e^{At}$, dominated by the cosmological constant.

(g) Big rip from phantom vacuum $\rho_V \propto R^\epsilon$

Problem. If $\rho_V \propto R^\epsilon$ with $\epsilon > 0$, show that $R(t)$ diverges in finite time in a flat universe (matter neglected after t_0).

Flat universe, ordinary matter neglected after t_0 . Friedmann equation reduces to

$$H^2 = \frac{8\pi G}{3} \rho_V.$$

Substitute $\rho_V = \rho_{V,0}(R/R_0)^\epsilon$:

$$H^2 = \frac{8\pi G}{3} \rho_{V,0} (R/R_0)^\epsilon.$$

Define $B^2 \equiv (8\pi G/3) \rho_{V,0} R_0^{-\epsilon}$:

$$H^2 = B^2 R^\epsilon.$$

Take positive root (expanding universe, $H > 0$):

$$H = B R^{\epsilon/2}.$$

Substitute $H = \dot{R}/R$:

$$\frac{\dot{R}}{R} = B R^{\epsilon/2}.$$

Rearrange (multiply by dt , divide by $R^{\epsilon/2}$):

$$R^{-1-\epsilon/2} dR = B dt.$$

Integrate from (t_0, R_0) to (t, R) :

$$\int_{R_0}^R R'^{-1-\epsilon/2} dR' = \int_{t_0}^t B dt'.$$

Evaluate (power rule, $\epsilon > 0$):

$$\left[\frac{R'^{-\epsilon/2}}{-\epsilon/2} \right]_{R_0}^R = B(t - t_0).$$

Simplify the prefactor:

$$-\frac{2}{\epsilon} \left[R^{-\epsilon/2} - R_0^{-\epsilon/2} \right] = B(t - t_0).$$

Solve for $R^{-\epsilon/2}$:

$$R^{-\epsilon/2} = R_0^{-\epsilon/2} - \frac{1}{2}\epsilon B(t - t_0).$$

$R \rightarrow \infty$ corresponds to $R^{-\epsilon/2} \rightarrow 0$:

$$0 = R_0^{-\epsilon/2} - \frac{1}{2}\epsilon B(t_{\text{rip}} - t_0).$$

Solve for $t_{\text{rip}} - t_0$:

$$t_{\text{rip}} - t_0 = \frac{2R_0^{-\epsilon/2}}{\epsilon B}.$$

Identify $H_0 \equiv H(t_0) = BR_0^{\epsilon/2}$, so $R_0^{-\epsilon/2} = B/H_0$:

$$t_{\text{rip}} - t_0 = \frac{2}{\epsilon H_0}.$$

Box the result:

$$\boxed{t_{\text{rip}} = t_0 + \frac{2}{\epsilon H_0} \quad (< \infty),}$$

finite for any $\epsilon > 0$ — the “big rip” singularity of phantom dark energy.